

MIDDLESEX UNIVERSITY LONDON

Faculty of Science and Technology

Department of Design Engineering and Mathematics

**GRASP FAILURE DETECTION AND RECOVERY IN ROBOTIC
ARM PICK-AND-PLACE SYSTEM**

BEng Mechatronics and Robotics

Major Project (PDE3823)

Technical Report

Myu Wai Shin

Student ID: M00964135

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Abstract:

Grasp failure in robotic pick-and-place systems remains a persistent operational problem, with reported failure rates ranging from 5 to 25 percent even in well-engineered systems. When failures go undetected, the robot continues as if the task succeeded, resulting in silent errors and wasted cycles. This project designs, implements, and evaluates a closed-loop grasp failure detection and recovery system on a UR10 robot arm with an OnRobot RG2 rigid parallel-jaw gripper, using only the limited proprioceptive signals available on this hardware class, combined with visual classification from a wrist-mounted eye-in-hand camera.

The system is a six-stage autonomous pick-and-place pipeline covering object detection, pose estimation, navigation, grasp execution, post-grasp verification, transit, and release. A two-layer failure detection architecture combines binary gripper contact and jaw-width signals with YOLO26n visual classification to determine grasp outcome at each stage. When failure is confirmed, a unified recovery routine re-localises the object and retries autonomously without human intervention.

Experimental evaluation demonstrates that the visual classifier achieves 92 percent accuracy and 100 percent recall on the holding class, exceeding the 90 percent target. The autonomous recovery routine achieves a 93 percent success rate, exceeding the 80 percent target. Navigation accuracy across a structured test workspace is 92 percent. All five project objectives were fully or partially met, demonstrating that reliable closed-loop failure detection and recovery are achievable on rigid parallel-jaw hardware without dedicated force-torque sensing or external camera infrastructure.

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Table of Contents

Abstract:	2
Acknowledgements.....	3
List of Figures	6
List of Tables.....	7
Chapter 1: Introduction	8
1.1 Background and Motivation	8
1.2 Problem Statement.....	8
1.3 Aims and Objectives	9
1.4 Scope and Contributions.....	11
Chapter 2: Technical Background and Literature Review	12
2.1 Introduction.....	12
2.2 Technical Background.....	12
2.2.1 Gripper Sensing Constraints and the False Positive Problem	12
2.2.2 Post-Grasp Verification: Why Direct Gripper Observation.....	12
2.2.3 Camera Configuration: Eye-in-Hand versus External	13
2.2.4 Slip Detection	13
2.3 Industry Solutions	14
2.4 Literature Review	15
2.5 Research Gap	15
Chapter 3: System Design and Implementation.....	16
3.1 System Overview	16
3.2 System Components	17
3.2.1 Hardware.....	17
3.2.2 Object Selection	19
3.2.3 Detection and Classification Models.....	19
3.2.4 Hand-Eye Calibration.....	21
3.2.5 Software Stack.....	22
3.3 Implementation Details.....	23

3.3.1 Stage 1: Explore.....	23
3.3.2 Stage 2: Navigation	24
3.3.3 Stage 3: Grasp	26
3.3.4 Stage 4: Verify	27
3.3.5 Stage 5 Transit.....	28
3.3.6 Stage 6: Release	28
3.3.7 Recovery.....	28
Chapter 4: Performance Evaluation Setup and Scenarios	29
4.1 Evaluation Methodology.....	29
4.2 Test Environment.....	30
4.3 Test Scenarios	31
4.3.1 Classifier Evaluation	31
4.3.2 Navigation Evaluation	31
4.3.3 Recovery Evaluation	32
Chapter 5: Results and Discussion.....	32
5.1 Experimental Results	32
5.1.1 Grasp Failure Detection — Visual Classifier	33
5.1.2 Navigation	34
5.1.3 Recovery.....	35
5.2 Performance Analysis	37
5.3 Discussion	38
5.4 Comparison with Related Work	39
Chapter 6: Conclusion and Future Work	40
6.1 Summary of Contributions	40
6.2 Limitations	40
6.3 Future Work.....	41
References	43
Appendix.....	45

List of Figures

Figure 1.2a: Pick-and-place trials conducted	9
Figure 2.3a: Amazon's Sparrow Robot	14
Figure 2.3b: RightHand Robotics' RightPick 4	14
Figure 3.1a: Full System Flowchart	16
Figure 3.2a: UR10 robot, OAK-D Lite camera, and RG2 gripper	17
Figure 3.2b: Custom PLA camera mount fitted on RG2 gripper body	18
Figure 3.2c: Technical drawing of the custom camera mount	18
Figure 3.2d: Detection training results	19
Figure 3.2e: Trained detection model on live inference	20
Figure 3.2f: Classifier validation results showing 1.0 accuracy on both classes	20
Figure 3.2g: Normalised confusion matrix showing no background confusion	21
Figure 3.2h: Eye-in-Hand camera to robot base transform	21
Figure 3.3a: Workspace test area enclosed by boundary markers.....	24
Figure 3.3b: Aligning the detected object to the frame centre	24
Figure 3.3c: Pose estimation using ArUco PnP in base frame (top left corner)	25
Figure 3.3d: Before centring correction (left) and after centring correction (right)	26
Figure 3.3e: False positive scenario in classification.....	28
Figure 4.2a: Test workspace layout	31
Figure 4.3a: Navigation test grid showing 12 tested positions across rows 2 and 3	32
Figure 5.1a: Classifier confusion matrix (n=50)	33
Figure 5.1b: Classifier confidence vs TCP-to-object distance	33
Figure 5.1c: Navigation test results showing hover alignment across 12 positions	35
Figure 5.1d: Recovery test results showing hover positions across 14 trials.....	36

List of Tables

Table 1	Comparison of related works and this project	9
Table 2	Software stack.....	17
Table 3	Robot communication ports	17
Table 4	Layer 1 detection logic	20
Table 5	Evaluation of metrics and targets per objective	23
Table 6	Classifier performance metrics	27
Table 7	Summary of results against objectives and targets	30

Chapter 1: Introduction

1.1 Background and Motivation

Robotic pick-and-place is one of the most widely deployed forms of industrial automation, found across warehouse fulfilment, manufacturing assembly, and logistics sorting. Advances in computer vision and deep learning have significantly improved how robotic systems detect objects, estimate pose, and plan grasps, enabling reliable handling across a much wider range of objects and environments than was previously possible.

Despite this progress, grasp failure remains a persistent operational problem. Reported failure rates in industrial bin-picking range from 5 to 25 percent depending on object geometry, surface conditions, and pose estimation accuracy [1]. Precision-picking is hard. A grasp that succeeds in planning can still fail at execution, and without a mechanism to detect it, the system continues as if the task succeeded. In high-throughput operations handling thousands of picks per shift, even low failure rates accumulate into a substantial number of undetected errors and wasted cycles.

Reliably determining whether a grasp has succeeded therefore becomes as important as planning the grasp itself. Research in robotic manipulation has shown that combining hardware feedback from the gripper with visual information produces more robust grasp state estimation than either source alone, as each modality covers failure modes the other cannot resolve independently. This multi-modal approach forms the foundation of effective grasp failure detection.

Rigid parallel-jaw grippers are among the most widely used end-effectors in industrial manipulation, capable of handling a broad range of object geometries. This project develops a closed-loop pipeline on this hardware class, fusing available gripper signals with visual information to detect grasp failure and execute autonomous recovery across the full pick-and-place cycle.

1.2 Problem Statement

Three categories of grasp failure were identified through preliminary experimentation. About 30 pick-and-place trials were conducted from known positions in a controlled, collision-free workspace.

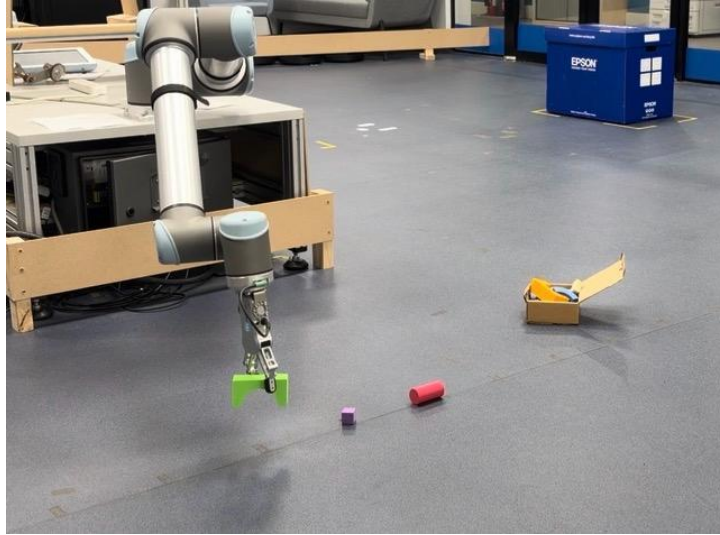


Figure 1.2a: Pick-and-place trials conducted

Object positions and orientations were deliberately varied between trials, and test objects of different shapes and sizes were used. This experiment revealed three distinct failures.

A pose error or miss occurs when depth-based localisation places the gripper off-target, causing it to close on empty air or make lateral contact without enclosing the object. In either case, the contact force limit is often reached because the jaws close against the floor or the object edge, producing sensor readings identical to a successful grasp. This is the fundamental false positive problem: gripper signals alone cannot distinguish a real grasp from a collision miss.

An orientation error occurs when the gripper approach axis is misaligned with the object geometry, preventing the jaws from enclosing it regardless of positional accuracy. In both pose and orientation failures, contact with the object during the failed attempt often displaces the target from its original position, so recovery cannot simply retry the same coordinates.

A slip failure occurs when the object is initially secured but lost during transit due to insufficient grip force or unstable contact geometry. These three failure modes are distinct in cause but share the property that they are not reliably detectable from gripper signals alone, motivating a multi-modal detection approach.

1.3 Aims and Objectives

The overall aim of this project is to design, implement, and experimentally evaluate a closed-loop grasp failure detection and recovery system on a robot with rigid parallel-jaw gripper, using sensor fusion between gripper feedback and camera-based visual classification.

Objective 1: Grasp Failure Detection

Implement a multi-modal failure detection pipeline that fuses gripper proprioceptive signals with visual classification to determine grasp outcome. The visual classifier should achieve an accuracy of $\geq 90\%$ on the grasp verification task.

Note: Nair et al. demonstrated 97% per-frame accuracy for image-based binary grasp verification using a camera mounted directly on a gripper [20]. The target of $\geq 90\%$ is set conservatively below this benchmark, reflecting that the classifier in this system must operate from a wrist-mounted eye-in-hand camera at varying clearance heights and object positions, rather than from a fixed external viewpoint under controlled conditions.

Objective 2: Autonomous Recovery

On confirmed failure, execute an autonomous recovery routine that re-detects the object at its current position and re-attempts the grasp without human intervention, retrying up to a defined maximum before aborting in irrecoverable scenarios. The target recovery success rate is $\geq 80\%$ across all confirmed failure cases.

Note: Morrison et al. report 83–88% grasp success rates in real-world closed-loop grasping trials using a wrist-mounted camera and parallel-jaw gripper [21]. The lower target of 80% accounts for the additional difficulty of recovering from a displaced object, requiring re-localisation before retrying, rather than re-grasping from a known workspace position.

Objective 3: Navigation

Navigate the robot TCP accurately to the detected object pose and execute a grasp. The system should achieve consistent hover positioning within the defined workspace.

Objective 4: Grasp Maximisation

Align and rotate the gripper to match the detected object's in-plane orientation and approach the object vertically from directly above to maximise contact stability. A top-down approach ensures both jaws engage the object symmetrically, which is the conventional and most reliable grasp configuration for a parallel-jaw gripper on a rigid object.

Objective 5: Slip Monitor (Extension Objective)

Detect object loss during the carry phase in near real-time and trigger recovery if the object is lost mid-transit. This objective is treated as an extension goal.

1.4 Scope and Contributions

The system is designed and evaluated for a single-object pick-and-place task in a structured, collision-free workspace. Testing is conducted on a single object class to allow controlled investigation of all three failure modes without introducing the additional complexity of multi-object detection or scene understanding. The scope is limited to detecting the predefined failure types described in Section 1.2, novel failures outside the training distribution are not addressed.

The following are the primary contributions of this work:

- A complete closed-loop pick-and-place pipeline with integrated failure detection and autonomous recovery, implemented and evaluated on real hardware without simulation.
- A multi-modal failure detection architecture that fuses binary gripper proprioception with camera-based visual classification, enabling reliable grasp state determination without continuous feedback from force-torque sensors or dedicated external camera infrastructure.
- An eye-in-hand camera configuration that provides a continuous view of both the workspace and the gripper state within a single fixed frame throughout the entire cycle, eliminating the need for a dedicated check pose and reducing cycle time.

The principal constraints and limitations of the project are as follows:

- The system is not designed to handle object occlusion, scene clutter, or generalisation to novel object classes. Extending the system to unstructured environments would require retraining the detection and classification models and is outside the scope of this project.
- The system is designed to detect a fixed set of predefined failure types. Novel failures not represented in the system (e.g., object deformation, stacking, or multiple overlapping objects) are outside the scope.
- Vision-based classification may fail under occlusion or poor lighting
- The system does not account for collision failure with the arm, gripper, or environment during motion.

Chapter 2: Technical Background and Literature Review

2.1 Introduction

This chapter provides the technical background underlying the key design decisions in this project and positions the work within the existing literature. It covers a review of relevant industry approaches and related academic work, and concludes with a comparison table and research gap analysis.

2.2 Technical Background

2.2.1 Gripper Sensing Constraints and the False Positive Problem

A successful parallel-jaw grasp requires the approach vector to align with the object geometry so both jaws make stable contact simultaneously [3]. In practice, accumulated pose estimation error or object displacement during descent contributes to grasp failure even when planning succeeds. The sensing available to detect these failures is hardware-dependent. Simulation tools such as GrasPlt! [4] have been widely used to evaluate grasp quality analytically, and data-driven planning approaches such as Dex-Net [15] demonstrate high grasp success rates using deep learning on synthetic point clouds. Both focus on improving planning rather than post-execution failure detection, which remains the less addressed problem.

More advanced rigid grippers expose continuous force signals through motor current sensing or integrated load cells. The OnRobot RG2 used in this project provides only a binary contact flag (DI8) and jaw-width feedback (AI2), representing the minimum available proprioception for this gripper class.

The consequence of this constraint is the false positive condition, where both the contact signal and jaw width report readings are consistent with a successful grasp, but the object has not been securely enclosed. This occurs when the jaws make lateral contact during descent, nudging the object sideways rather than grasping it [2]. Since a nudge and a genuine grasp produce identical sensor readings, the ambiguity cannot be resolved using gripper signals alone. This is the primary motivation for incorporating camera-based visual classification as the definitive grasp state confirmation method in this project.

2.2.2 Post-Grasp Verification: Why Direct Gripper Observation

Camera-based approaches to grasp verification span several methods reviewed in Du et al. [5], ranging from workspace state checking to direct gripper observation. Three broad approaches to

camera-based grasp verification were observed in the literature. The first is workspace state checking: inferring success from the absence of the object after pick. The second is object tracking: an external camera tracks object position throughout the approach and lift, inferring success if the object moves upward with the gripper. Zhu et al. [2] use this method, but the robot arm occludes the object during floor-level descent, making tracking unreliable at the moment of contact.

The third approach, adopted in this project, is direct observation of the gripper jaws after the lift. A classifier determines whether an object is present between the jaws from a single frame captured at a fixed clearance height. Amargant et al. [6] demonstrate that this approach achieves high verification accuracy using a two-stage YOLO plus ResNet pipeline. This method directly answers whether a grasp succeeded rather than inferring it indirectly, and is robust to the occlusion problem that limits object tracking approaches.

2.2.3 Camera Configuration: Eye-in-Hand versus External

Direct gripper jaw observation requires the camera to have a clear view of the gripper jaws after the lift. Two configurations are used in robotic arm picking: an external fixed camera and an eye-in-hand camera that moves with the robot.

An external fixed camera requires the robot to move to a dedicated check pose after every grasp attempt, adding cycle time. Amargant et al. [6] use a head-mounted camera, but their architecture still requires an explicit gripper localisation step, as the gripper is not guaranteed to be in a consistent part of the frame.

The conventional wrist-mount orientation aligns the camera parallel to the approach axis, giving a forward-facing view from the gripper's perspective rather than a view of the gripper itself. With this configuration, the jaws are not visible, so grasp verification requires an additional wrist rotation to bring the jaws into the camera frame.

In this project, the camera is mounted at a downward angle on the gripper body, so that both the workspace below and the gripper jaws are simultaneously visible within the same frame throughout the full execution cycle. This eliminates the need for a dedicated check pose and supports object detection and post-grasp verification from a single fixed camera configuration.

2.2.4 Slip Detection

Slip during transit occurs when an object initially secured in the gripper is lost due to insufficient grip force or unstable contact geometry. Li et al. [10] demonstrate that combining tactile and visual information achieves 88% slip detection accuracy, highlighting that multi-modal sensing outperforms either modality alone.

Tactile sensor integration was not pursued in this project given the complexity of the full pipeline implementation and the time constraints of completing all components. Slip detection is therefore implemented using visual information only, and its performance is evaluated as part of the system testing in Chapter 5.

2.3 Industry Solutions

Amazon's Sparrow robot [16] uses computer vision and AI to detect and select individual products from unstructured scenes, combining camera data with prior knowledge of object structure to plan and execute grasps across millions of product types.

Earlier work from the Amazon Picking Challenge, reviewed in Eppner et al. [7], highlighted that reliable object localisation and grasp execution in unstructured bin environments remains a hard problem even at industrial scale, reinforcing that failure handling cannot be fully solved at the planning stage alone.



Figure 2.3a: Amazon's Sparrow Robot

RightHand Robotics' RightPick 4 system [17] follows a similar approach and additionally incorporates smart grasp and suction seal quality detection as part of its picking pipeline, reflecting the industry recognition that verifying grasp outcome is a necessary part of reliable autonomous operation.

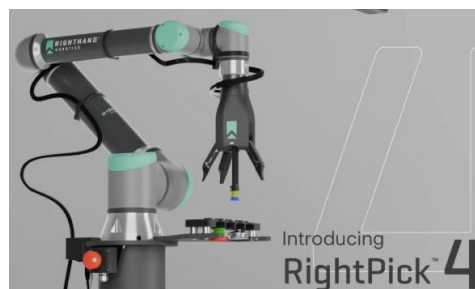


Figure 2.3b: RightHand Robotics' RightPick 4

Both systems show that combining object detection, grasp planning, and post-pick verification within a single automated pipeline is a proven approach at an industrial scale. This project applies the same pipeline structure to a rigid parallel-jaw gripper platform, extending post-pick

verification to direct visual observation of the gripper state and replacing exception flagging with a full autonomous recovery loop that re-localises the object and retries without human intervention.

2.4 Literature Review

The following comparison evaluates directly related works alongside this project across key criteria relevant to grasp failure detection and recovery. Zhu et al. are included for their closed-loop failure handling across the full pick-and-place cycle, and Amargant et al. for their camera-based binary grasp verification approach. Both works directly inform the design decisions in this project, and are evaluated and compared in Table 1 below.

Criterion	Zhu et al. [2]	Amargant et al. [6]	This project
End-effector	Pneumatic soft gripper	Unspecified rigid gripper	OnRobot RG2 parallel-jaw
Grasp sensing	Continuous internal bellow pressure	Binary gripper state only	contact signal + jaw width
Verification method	Visual object tracking across all phases	gripper localisation + ResNet classification	direct gripper jaw observation
Object detection	Custom visual tracker	YOLO detection	YOLO detection
Camera config	Fixed external camera	Head-mounted camera	Wrist-mounted camera at 21° downward tilt
Slip detection	Pressure signal drop during transit	Not addressed	Live visual gripper monitoring during transit
Recovery	Full state machine, type-specific	Not addressed	Unified re-detect and retry
Key limitation	Requires pneumatic gripper	Detection only, no recovery	Sensor ambiguity between contact failure subtypes

Table 1: Comparison of related works and this project

2.5 Research Gap

The literature identifies two partial solutions to the grasp failure problem. Zhu et al. [2] demonstrate closed-loop failure handling across all pick-and-place phases but rely on pneumatic

soft gripper pressure sensing, which has no equivalent on rigid parallel-jaw hardware. Amargant et al. [6] demonstrate camera-based binary grasp verification at high accuracy but address detection only, with no recovery pipeline and a camera configuration that requires an explicit gripper localisation step before classification can run.

No published work combines closed-loop failure detection and autonomous recovery on a rigid parallel-jaw gripper using only the limited signals available on this hardware class. This project fills that gap by fusing gripper proprioceptive feedback with visual information from a wrist-mounted eye-in-hand camera into a full autonomous pick-and-place pipeline with integrated failure detection and recovery.

Chapter 3: System Design and Implementation

3.1 System Overview

The system consists of a six-stage autonomous pick-and-place pipeline with integrated grasp failure detection and recovery. The stages are executed sequentially: Explore, Navigate, Grasp, Verify, Transit, and Release. When failure is confirmed at any of these points, a unified recovery routine relocates the object and returns control to Stage 2 for a fresh attempt. The pipeline aborts if the total failure count across all attempts reaches five. The full pipeline flow is shown in Figure 3.1a. The methodology and design of each step are covered in Section 3.3 Implementation Details.

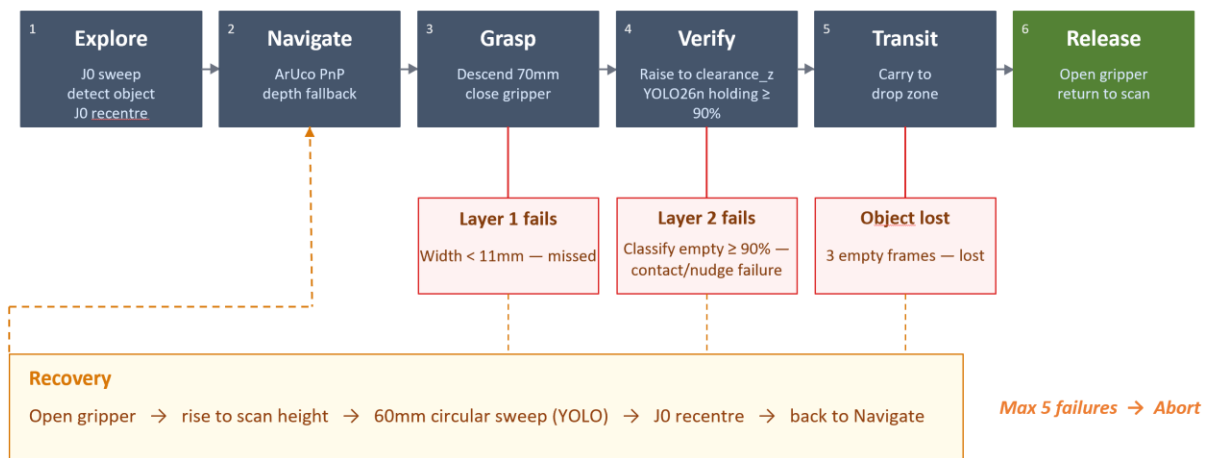


Figure 3.1a: Full System Flowchart

All stages are orchestrated by a top-level `main.py` script that sequences the pipeline, passes state between stages, handles failure events, and manages the retry counter.

Failure detection operates at three points in the pipeline. At Stage 3, gripper jaw-width feedback provides an immediate hardware-level check immediately after the gripper closes. At Stage 4, a visual classifier determines whether the object is present between the jaws after lifting to a fixed clearance height. During Stage 5, the same classifier runs continuously on live camera frames throughout transit to detect object loss mid-carry. These three detection points correspond to the three failure modes identified in Section 1.2: a complete miss, a contact failure where the grip is not secure, and a slip during transport.

3.2 System Components

3.2.1 Hardware



Figure 3.2a: UR10 robot, OAK-D Lite camera, and RG2 gripper

The robot platform is a Universal Robots UR10 CB-Series 6-DOF collaborative arm with a 1,300 mm reach and 10 kg payload capacity [11]. The large workspace coverage makes it well-suited for the pick-and-place task.

The end-effector is an OnRobot RG2 parallel-jaw gripper with a maximum jaw span of 110 mm and a 2 kg payload. The RG2 exposes two signals relevant to failure detection: a binary contact flag on digital input DI8, which goes high when the gripper detects a force contact, and a continuous analogue voltage on AI2 that is converted to jaw-width in millimetres using the conversion specified in the RG2 datasheet [12] (*see Appendix E*). These two signals form the basis of the Layer 1 grasp failure check described in Section 3.3.3 (Step 3: Grasp).

The perception sensor is an Intel OAK-D Lite stereo RGB-D camera [13]. It provides a 1280x720 RGB stream for object detection and classification, and a calibrated stereo depth map aligned to the RGB frame for 3D localisation of markerless objects. The camera connects over USB and runs via DepthAI Python SDK.



Figure 3.2b: Custom PLA camera mount fitted on RG2 gripper body

The camera is mounted on a custom 3D-printed PLA bracket attached to the body of the RG2 gripper. The mount positions the camera slightly behind the TCP and tilts it 21 degrees downward. This angle was chosen so that both the gripper jaws and the workspace directly below are simultaneously visible in the same frame throughout the execution cycle, without requiring any change of pose between the object detection phase and the post-grasp verification phase.

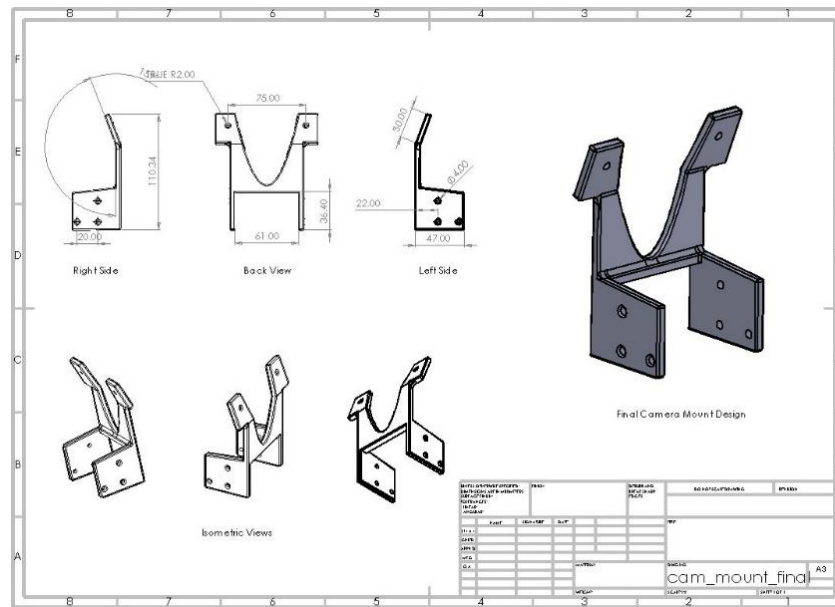


Figure 3.2c: Technical drawing of the custom camera mount

3.2.2 Object Selection

The target object is a vitamin bottle selected for two specific reasons relevant to this project. First, its cylindrical body makes it prone to slip because the curved surface provides less stable jaw contact than a flat-sided object, making it a suitable object for demonstrating objective 5, slip detection. Second, the cylindrical shape means the bottle can land in different orientations when placed or displaced, allowing objective 4, grasp maximisation to be demonstrated, where the system must detect the orientation and rotate the gripper accordingly to achieve a stable grasp.

An ArUco marker is affixed to the base face of the bottle. When the base face is upward and the marker is visible to the camera, ArUco-based pose estimation is used for navigation. When the marker is not visible due to the bottle being in a different orientation, the system falls back to markerless depth-based estimation.

3.2.3 Detection and Classification Models

Visual AI is used at two points in the pipeline: detection and classification. Both tasks use YOLO26n [19], which is selected because:

- It operates reliably from scan height with less background confusion than earlier YOLO versions
- It is lightweight for CPU inference without a dedicated GPU
- It is NMS-free at deployment, so no post-processing threshold tuning is needed.

The detection model is used in Stage 1 (*refer to Section 3.1*) to localise the object during the workspace sweep, and during Stage 2 as the fallback bounding box source for depth-based pose estimation (*see Section 3.3.2*).

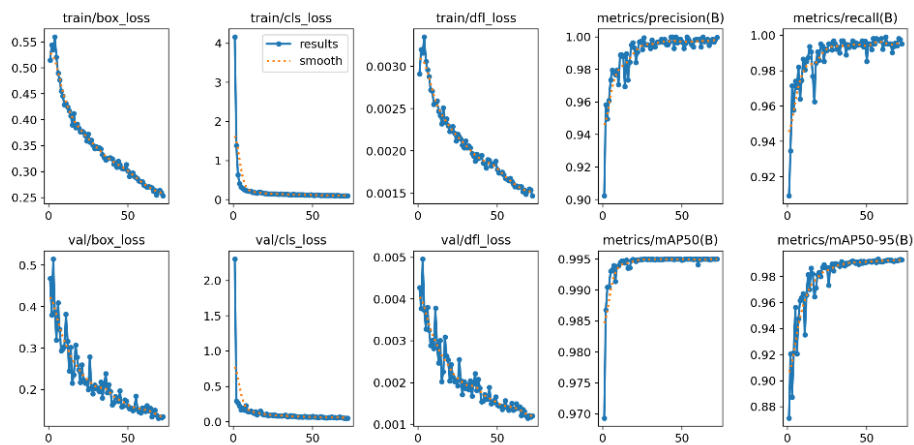


Figure 3.2d: Detection training results

The model achieves mAP@0.5 of 0.995 on the validation set with all three loss curves converging cleanly. On live frames, it maintains 94 to 98 percent confidence consistently with no false positives above the 0.80 threshold.



Figure 3.2e: Trained detection model on live inference

The classification model is used during Stage 4 (*refer to Section 3.1*) to verify the grasp outcome after lifting to clearance height, and runs continuously during Stage 5 to monitor for object loss during transit. Vision-language models such as CLIP [9] were evaluated for object classification during development but replaced by YOLO26n after achieving lower accuracy on the grasp verification task with the same dataset. Trained on 7,200 images split equally between the holding and empty classes at clearance heights between 200 mm and 600 mm. The model reaches top-1 accuracy of 1.0 on the validation set, with both empty and holding classes achieving 1.0 on the normalised confusion matrix.

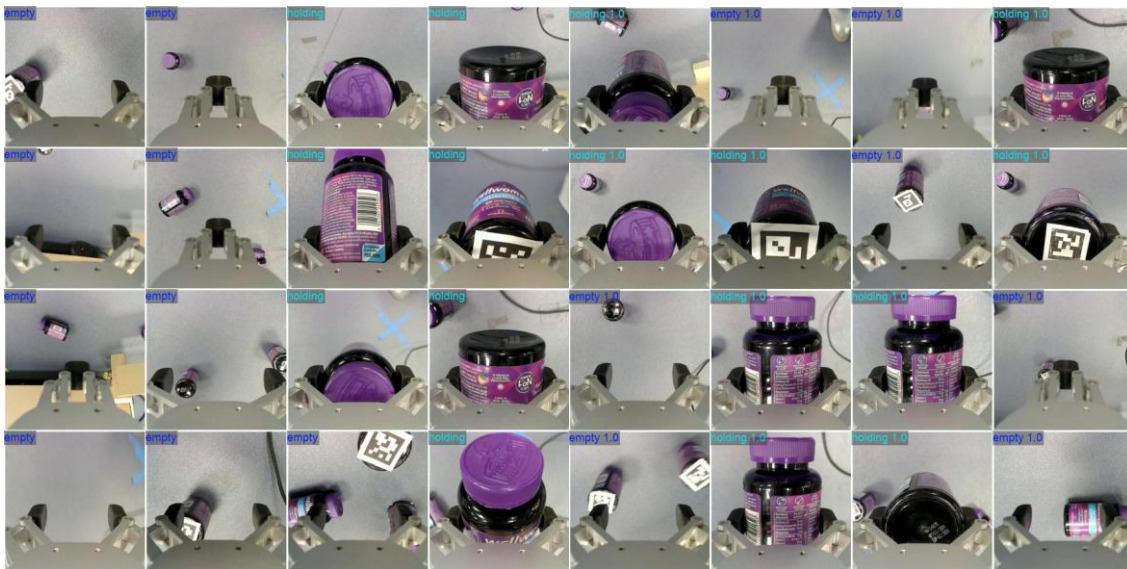


Figure 3.2f: Classifier validation results showing 1.0 accuracy on both classes

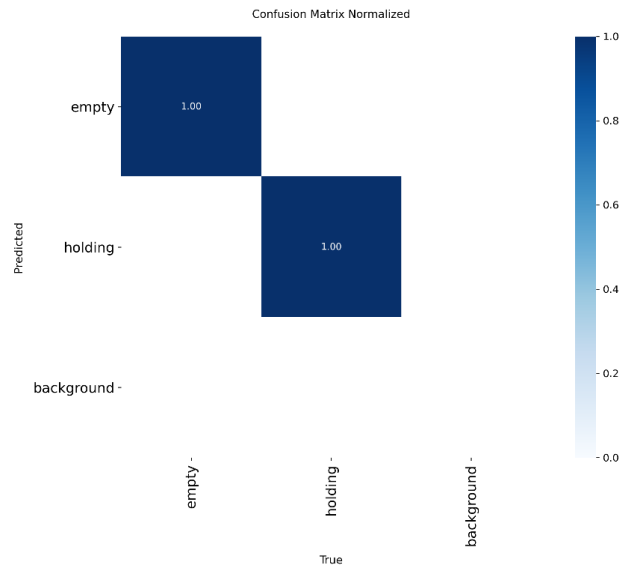


Figure 3.2g: Normalised confusion matrix showing no background confusion

3.2.4 Hand-Eye Calibration

Hand-eye calibration determines the rigid transform from the camera frame to the robot flange frame, which is then combined with the live robot forward kinematics to map any point observed in the camera into the robot base frame. The full transform chain is visualized in the diagram below.

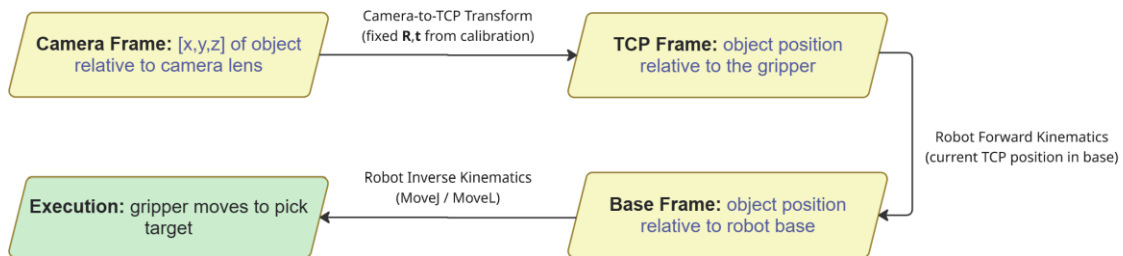


Figure 3.2h: Eye-in-Hand camera to robot base transform

The equation:

$$T_{\text{tag2base}} = T_{\text{tcp2base}} \times T_{\text{cam2tcp}} \times T_{\text{tag2cam}}$$

- `T_tag2cam` is computed per-frame by ArUco solvePnP using the known marker size.
- `T_cam2tcp` is the calibrated hand-eye transform, loaded at startup from `T_cam2tcp.npy`.
- `T_tcp2base` is read live from the robot forward kinematics at each frame.

The resulting `T_tag2base` gives the tag position and orientation directly in robot world coordinates, which becomes the hover target for the TCP.

The extrinsic hand-eye transform `T_cam2tcp` was estimated using the Tsai-Lenz method [8] via OpenCV's `calibrateHandEye()` function. Calibration data was collected across 30 robot configurations with deliberate variation in wrist orientation, lateral position across the workspace, and height from 100 to 900 mm above the floor, with robot TCP poses recorded via the `ur_rtde` interface [14]. The Tsai method solves the $AX=XB$ formulation, where A represents the relative motion between robot poses, B represents the corresponding relative motion observed by the camera, and X is the unknown fixed transform from the camera to the TCP, the rigid offset and rotation that remains constant regardless of the arm configuration. Solving across all collected pose pairs yields the calibrated `T_cam2tcp`.

3.2.5 Software Stack

Layer	Tool
Robot communication	Python socket/URScript
Computer vision	OpenCV
Camera	DepthAI
Detection and classification	Ultralytics YOLO

Table 2: Software

The software stack is implemented in Python across four functional layers. Robot communication is handled via raw Python sockets, sending URScript commands directly to the controller. Computer vision operations, including ArUco detection, pose estimation, and the hand-eye transform are handled by OpenCV [18]. Camera streaming and stereo depth are accessed through the DepthAI SDK. Detection and classification inference runs through the Ultralytics YOLO framework.

Port	Purpose
30002 (write)	Motion commands sent as URScript
30002 (read)	Reads robot state: TCP pose, joints, gripper width
29999	Gripper control via URP programs

Table 3: Robot Communication Ports

Robot communication runs over two TCP socket connections. Port 30002 is the primary channel for sending URScript motion commands and reading back robot state data, including TCP pose, joint angles, and gripper signals continuously. Port 29999 is the dashboard server, used only to load and trigger URP programs for gripper operation.

Multithreading is used throughout the pipeline so that robot state polling, camera inference, and motion execution run in parallel without blocking each other, ensuring the system always has current state data and live classifier output regardless of what motion is executing.

3.3 Implementation Details

3.3.1 Stage 1: Explore

The explore stage is responsible for locating the object in the workspace and centring it in the camera frame before pose estimation begins. The robot first moves to a pre-recorded scan pose that positions the camera with a clear overhead view of the full workspace. Once there, the base joint J0 sweeps across the workspace from -0.5 to +0.5 radians at 0.2 rad/s while YOLO26n detection runs continuously on live camera frames. J0 is used for the sweep because it covers the widest horizontal area per radian of movement compared to any other joint. As soon as a detection fires above the 0.80 confidence threshold, the sweep stops immediately.

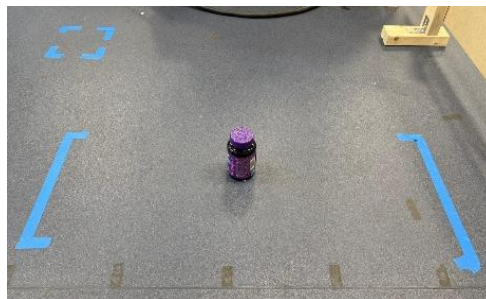


Figure 3.3a: Workspace test area enclosed by boundary markers

If no object is found after two full sweeps, the arm descends 100 mm to a lower scan height and repeats the sweep, giving the camera a closer view of the workspace in case the object was not detectable from the initial height.



Figure 3.3b: Aligning the detected object to the frame centre

Once the object is detected, an active recentering loop runs before handing off to navigation. JO is adjusted in up to six proportional correction steps until the object pixel centre is within 20 pixels of the image centre. This step matters because pose estimation accuracy degrades toward the image periphery due to lens distortion and perspective error, and centring the object first ensures the most accurate possible pose estimate in the next stage.

3.3.2 Stage 2: Navigation

The navigate stage computes the object's 3D position and orientation in the robot base frame, moves the TCP to a hover position directly above the object, and aligns the gripper before descending. Three things happen in sequence: pose estimation, hover move with centring correction, and gripper orientation alignment.

Pose Estimation

Two methods are available depending on what the camera can see. The primary method uses ArUco PnP. The known marker on the base face is detected by the RGB camera, and solvePnP recovers the full 6-DOF pose from the four tag corners, giving both 3D position and in-plane

orientation. This is the preferred method because it does not depend on stereo depth and provides the orientation needed for gripper alignment as a proof of concept for maximising grasp.

The fallback method is used when the marker is not visible. The YOLO26n bounding box centre provides the XY pixel position, and the OAK-D stereo depth at that pixel provides the Z distance. These are back-projected into 3D camera coordinates using the camera intrinsics and transformed into the robot base frame. This gives position only with no orientation. Navigation using this method was not fully reliable due to an unresolved error in the transformation chain that was not debugged within the project timeline, and it is therefore included as a partial implementation.



Figure 3.3c: Pose estimation using ArUco PnP in base frame (top left corner)

Hover Move, Centring Correction and Gripper Alignment

Before the hover move is triggered, the estimated target must remain stable within 5 mm across five consecutive frames to filter single-frame noise. After extracting the pose estimation, the robot executes a linear Cartesian move to the hover position. The robot moves to the hover position while simultaneously rotating the gripper to match the detected tag orientation, aligning the jaw axis to the object, and also ensuring the TCP Z-axis points straight down toward the floor on arrival.

After arrival, a post-hover centring correction loop applies up to three small horizontal TCP corrections to reduce any residual calibration offset. This correction is possible because the camera mount angle keeps the object visible in the frame even at hover height without occlusion from the gripper body, allowing the pixel misalignment to be measured and corrected.

When ArUco was used for pose estimation, PnP is re-run on a fresh frame, and the pixel offset is converted to metres using the PnP-derived depth and focal length. When the depth fallback was used, the stereo depth at the updated bounding box centre is used for the same correction. In both cases the correction moves the TCP horizontally until the object is within the centre tolerance of the frame.



Figure 3.3d: Before centring correction (left) showing a slight TCP offset from the tag centre, and after centring correction (right) showing TCP aligned above the tag centre

Centring correction, gripper yaw alignment, and vertical TCP orientation all collectively form the grasp maximisation strategy for Objective 4. Pose estimation alone is not sufficient to guarantee a successful grasp. Residual calibration error, depth noise, and lens distortion all introduce offsets between the estimated hover position and the true object centre. Correcting for these before descent maximises the probability of a stable, symmetric grasp on every attempt.

3.3.3 Stage 3: Grasp

With the TCP at hover height directly above the object, the robot descends 70 mm and closes the gripper via a URP program triggered through the dashboard port. The 70 mm descent was chosen to ensure sufficient jaw contact surface on the bottle body, accounting for the 33 mm cushion length and the offset between the hover position and the physical contact surface. This value was validated through testing.

Immediately after the gripper closes, Layer 1 detection evaluates the two gripper signals. The DI8 contact flag indicates whether the force limit was reached during closing. The AI2 jaw-width reading indicates whether something of sufficient size is between the jaws. The 11 mm threshold is the decision boundary for Layer 1: a reading below it means the jaws have closed fully with nothing between them, while a reading above it indicates something is preventing full closure. All combinations of these two signals and their outcomes are shown in Table 4.

DI8 Contact	AI2 Jaw Width	Interpretation	Action
LOW (no contact)	≤ 11 mm	Miss (gripper closed on air)	→ Recover
LOW (no contact)	> 11 mm	Partial contact/object nudged	→ Recover
HIGH (contact)	≤ 11 mm	Contact but fully closed (object lost)	→ Recover
HIGH (contact)	> 11 mm	Object between jaws	→ Stage 4 Verify

Table 4: Layer 1 detection logic: gripper signal combinations and resulting actions

All combinations other than HIGH contact with jaw width above 11 mm trigger recovery immediately. The single passing combination does not confirm a successful grasp but only senses that something may be between the jaws. A false positive is still possible here, where the gripper made contact with the object edge without actually enclosing it, and the sensor readings look identical to a real grasp. This is why all cases that pass Layer 1 are forwarded to Stage 4 for visual confirmation.

3.3.4 Stage 4: Verify

After picking the robot lifts to a clearance height of 400 mm above the hover position to verify grasp success. At this height, the wrist-mounted camera has a clear overhead view of the closed gripper. The YOLO26n classifier evaluates a single camera frame and returns a confidence score for the holding class. If the holding confidence is at or above 0.90, the grasp is confirmed, and transit begins. If the classifier returns empty or confidence falls below the threshold, a contact failure is confirmed, and recovery is triggered.

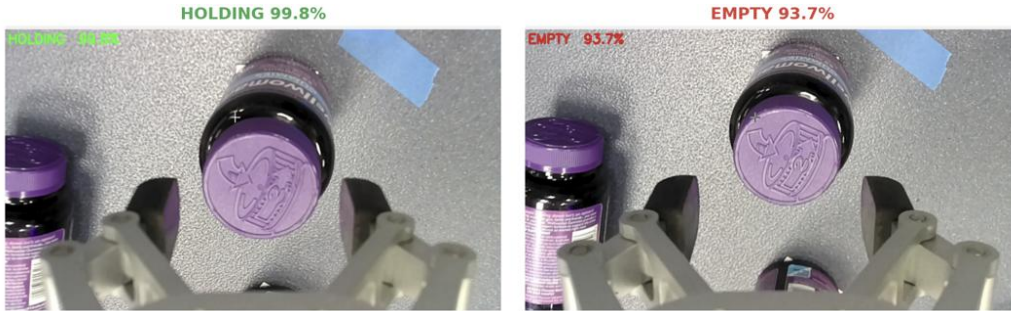


Figure 3.3e: False positive scenario in classification

The 400 mm clearance height was determined through classifier testing. At close range, below approximately 150 mm TCP-to-object distance, the bottle can appear between the jaws in the camera frame even when not physically grasped, causing the classifier to incorrectly predict holding. Lifting to 400 mm eliminates this false positive condition entirely, and the classifier operates at near 100 percent confidence consistently at this height. The clearance height therefore serves two purposes: it is the point at which visual verification is performed, and it is the operating height maintained throughout transit.

3.3.5 Stage 5 Transit

Once the grasp is confirmed, the robot carries the object horizontally to a fixed drop zone at clearance height, with the drop zone coordinates loaded from a pre-recorded file. Throughout the move, a background thread runs the YOLO26n classifier continuously on live camera frames to monitor for object loss. Three consecutive empty classifications below the 0.90 confidence threshold are required to trigger a stop and initiate recovery. The three-frame requirement prevents false triggers from isolated blurry or motion-affected frames that can occur during arm movement.

3.3.6 Stage 6: Release

On reaching the drop zone, the robot descends by the same distance it rose after the grasp, ensuring the object is placed at the correct height above the drop surface. The gripper opens via a URP program, the arm rises back to clearance height to clear the drop zone, then returns to the scan pose ready for the next cycle.

3.3.7 Recovery

Recovery is triggered by a Layer 1 failure, a Layer 2 failure, or a transit slip detection. The same routine runs regardless of which stage triggered it. The gripper opens immediately to release any partial contact, and the arm rises to scan height above the area where the object was last seen. A 60 mm radius circular sweep, chosen to cover the expected displacement range after a failed grasp, then runs across five waypoints with YOLO26n detection active on every frame. As soon as a detection fires, the sweep stops and the J0 recentring loop adjusts until the object is within 20

pixels of the frame centre, mirroring the same recentring step used in Stage 1. Control then returns to Stage 2 for a fresh pose estimation and hover attempt. The failure counter increments on every recovery entry, and the pipeline aborts after five total failures, a limit chosen to handle transient failures while preventing indefinite looping in irrecoverable scenarios.

Chapter 4: Performance Evaluation Setup and Scenarios

4.1 Evaluation Methodology

Objectives defined in Section 1.3 are evaluated against a specific performance metric. Each metric was selected to measure whether the system achieved what it set out to do in practice. Table 5 summarises the metrics, targets, and justification for each objective.

Objective	Metric	Target	Justification
Obj 1: Grasp Failure Detection	Classification accuracy, recall	$\geq 90\%$ accuracy	Accuracy reflects overall correctness; recall ensures no real grasp is missed
Obj 2: Autonomous Recovery	Recovery success rate	$\geq 80\%$	Measures whether the system completes the task after a confirmed failure
Obj 3: Navigation	Hover alignment rate	Consistent across workspace	Measures whether the TCP lands within the graspable zone above the object
Obj 4: Grasp Maximisation	Qualitative assessment	Proof of concept	Gripper aligns to tag orientation and approaches vertically
Obj 5: Slip Detection (Extension)	Detection during transit	Detects object loss	Verified through live classifier monitoring during transit

Table 5: Evaluation of metrics and targets per objective

For Objective 1, classification accuracy measures the proportion of correctly classified frames across both holding and empty classes. Recall is additionally tracked because a false negative,

classifying a real grasp as empty, would cause the system to incorrectly trigger recovery on a successful pick, making it the more critical error type to avoid.

For Objective 2, the recovery success rate is defined as the proportion of trials in which the system successfully re-navigated to a hover position above the object and completed a grasp attempt within the maximum retry limit. This directly measures whether the recovery routine fulfils its purpose of completing the task autonomously after failure.

For Objective 3, navigation accuracy is defined as the proportion of hover positions that result in a successful grasp when the gripper is closed at that position. After hovering, the gripper was closed, and the outcome recorded: a pick is considered aligned if the object was successfully enclosed and lifted, and not aligned if the gripper missed or failed to secure the object.

Objectives 4 and 5 are evaluated qualitatively. Grasp maximisation is assessed by verifying that the gripper rotates to the detected orientation and descends vertically. Slip detection is assessed by verifying that object loss during transit is detected, and recovery is triggered correctly.

4.2 Test Environment

All testing was conducted on the physical robot platform described in Section 3.2, with no simulation used at any stage. The hardware configuration during testing was identical to that used during development.

The test workspace was defined by a 4x6 grid of positions at 150 mm pitch, laid out using four A3 sheets arranged in a 2x2 configuration on the floor within the robot's reachable area. The origin was fixed at the bottom-left corner of the layout. Rows 1 and 4 were outside the camera sweep view and were not tested. Testing was conducted on rows 2 and 3, giving a 6x2 grid of 12 positions within the effective workspace.

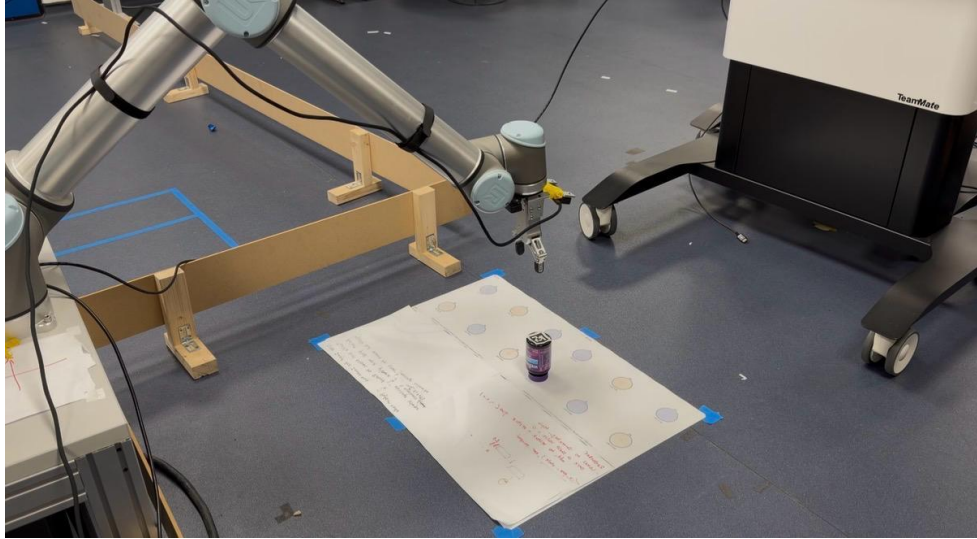


Figure 4.2a: Test workspace layout showing 4x6 grid with rows 2 and 3 highlighted as the tested area

4.3 Test Scenarios

4.3.1 Classifier Evaluation

The visual classifier was evaluated on a controlled set of 50 trials conducted at clearance height. Each trial presented the gripper in either the holding or empty state, with the ground truth known beforehand. The holding trials covered the bottle gripped at different orientations between the jaws. The empty trials included different jaw widths, objects visible between the jaws but not physically grasped, and partial occlusion scenarios to represent the false positive condition. Height was also varied across trials to characterise how classifier confidence changes with TCP-to-object distance. Results were recorded as a confusion matrix across the two classes.

4.3.2 Navigation Evaluation

Navigation accuracy was evaluated across 12 positions in a 6x2 grid at 150 mm pitch, covering rows 2 and 3 of the test workspace. Only ArUco PnP-based navigation was evaluated in this test. The depth-based fallback method was not tested independently as it was not sufficiently reliable for navigation within the project timeline. For each trial, the full explore and navigate pipeline was run from the scan pose with the ArUco marker facing upward. After hovering, the gripper was closed and the object lifted to determine whether navigation had placed the TCP accurately enough for a successful grasp. A pick is considered aligned if the object was enclosed and lifted, and not aligned if the gripper missed.



Figure 4.3a: Navigation test grid showing 12 tested positions across rows 2 and 3

4.3.3 Recovery Evaluation

Recovery was evaluated across 14 valid trials at randomly scattered positions within the controlled workspace. The bottle was placed at an arbitrary position without recording the exact placement coordinates. The recovery routine was called directly for each trial, executing the 60 mm circular search sweep, re-detecting the object, and re-navigating to a hover position above it. The final hover TCP coordinates and recovery outcome were logged per attempt. Camera hardware disconnects errors occurring during the search phase were excluded as they represent hardware faults rather than algorithmic failures, leaving 14 valid trials with successful ArUco detections for evaluation. Each trial was allowed up to five attempts before being marked as a failure.

Chapter 5: Results and Discussion

5.1 Experimental Results

This section presents the experimental results across three evaluated components: visual classifier performance, navigation accuracy, and recovery success rate. Grasp maximisation is assessed qualitatively and slip detection is verified through pipeline behaviour rather than independent testing.

5.1.1 Grasp Failure Detection — Visual Classifier

The YOLO26n binary classifier was evaluated on 50 trials at clearance height. Results are summarised in Table 6, and the confusion matrix is shown in Figure 5.1a.

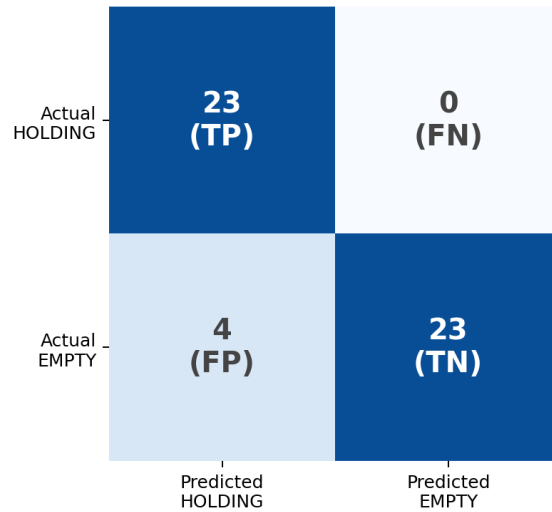


Figure 5.1a: Classifier confusion matrix (n=50)

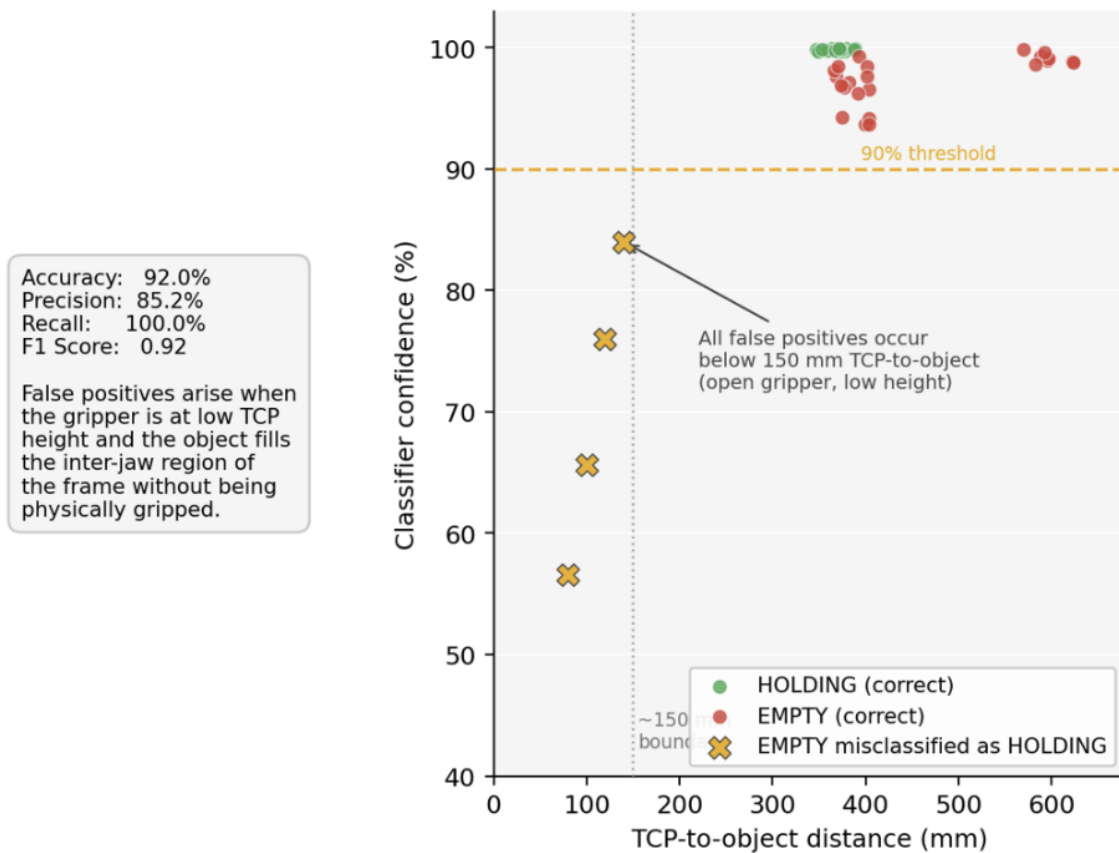


Figure 5.1b: Classifier confidence vs TCP-to-object distance showing all false positives occur below 150 mm

Metric	Value	Target
Accuracy	92%	≥ 90%
Recall (holding class)	100%	Maximise
Precision	85.2%	-
F1 Score	0.92	-
False positives (FP)	4	0 above 150 mm
False negatives (FN)	0	0

Table 6: Classifier performance metrics

The classifier achieved 92 percent overall accuracy and 100 percent recall on the holding class, meaning no genuine grasp was misclassified as empty across all 50 trials. The four false positives occurred exclusively below 150 mm TCP-to-object distance, visible in Figure 5.1b. No false positives were observed above 150 mm. Since the pipeline always classifies at a 400 mm clearance height, the effective false positive rate in operation is zero.

5.1.2 Navigation

Navigation was evaluated across 12 positions in a 6x2 grid at 150 mm pitch. The system achieved successful hover alignment in 11 of 12 positions, giving a navigation accuracy of 92 percent.

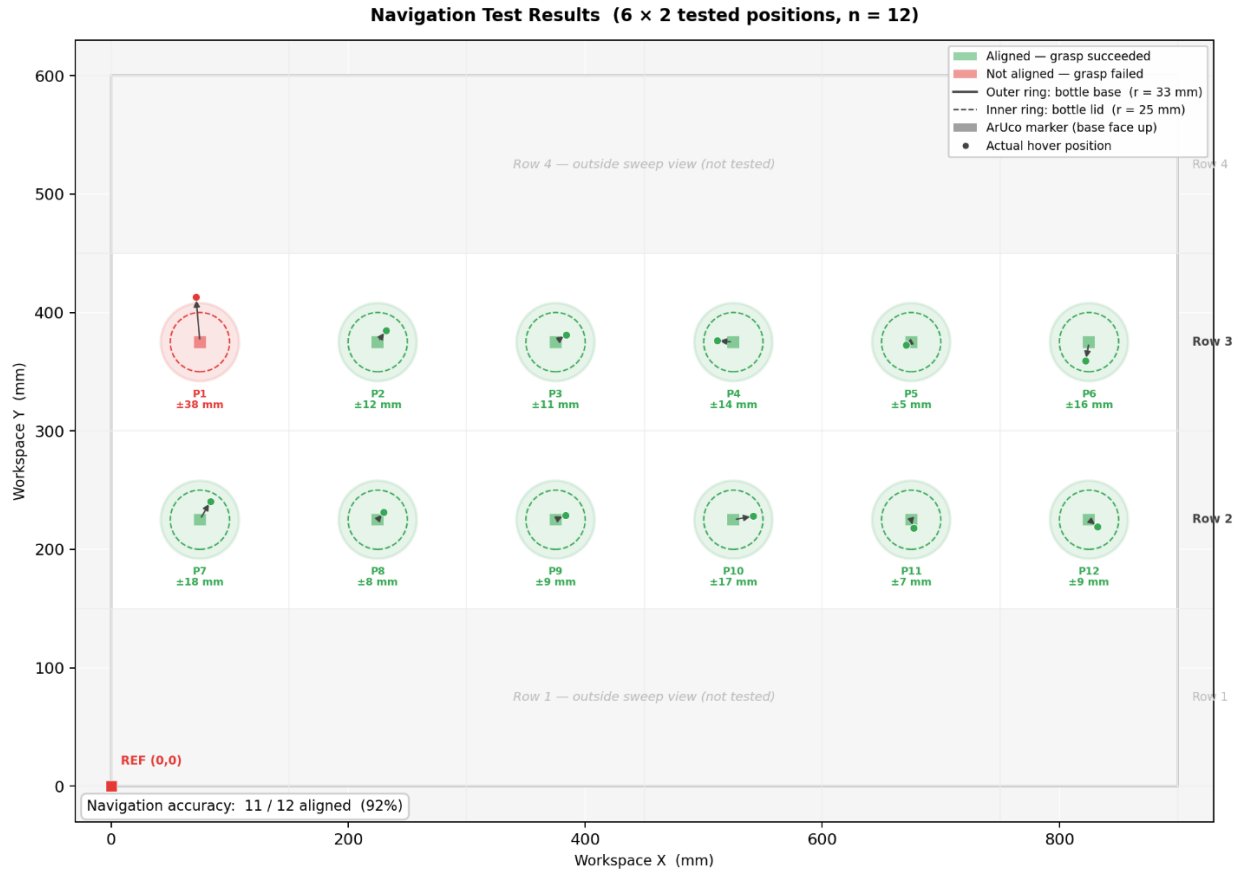


Figure 5.1c: Navigation test results showing hover alignment across 12 positions

The single failure occurred at P1, the leftmost position in row 3. This position is at the far edge of the tested workspace and produced the largest hover offset, and prevented a successful grasp. All other positions across both rows produced accurate hover positions within the graspable zone. No systematic degradation was observed across the width of the workspace within the tested region.

5.1.3 Recovery

Recovery was evaluated across 14 valid trials at randomly scattered positions within the controlled workspace. The system successfully re-navigated to a hover position above the object in 13 of 14 trials, giving a recovery success rate of 93 percent.

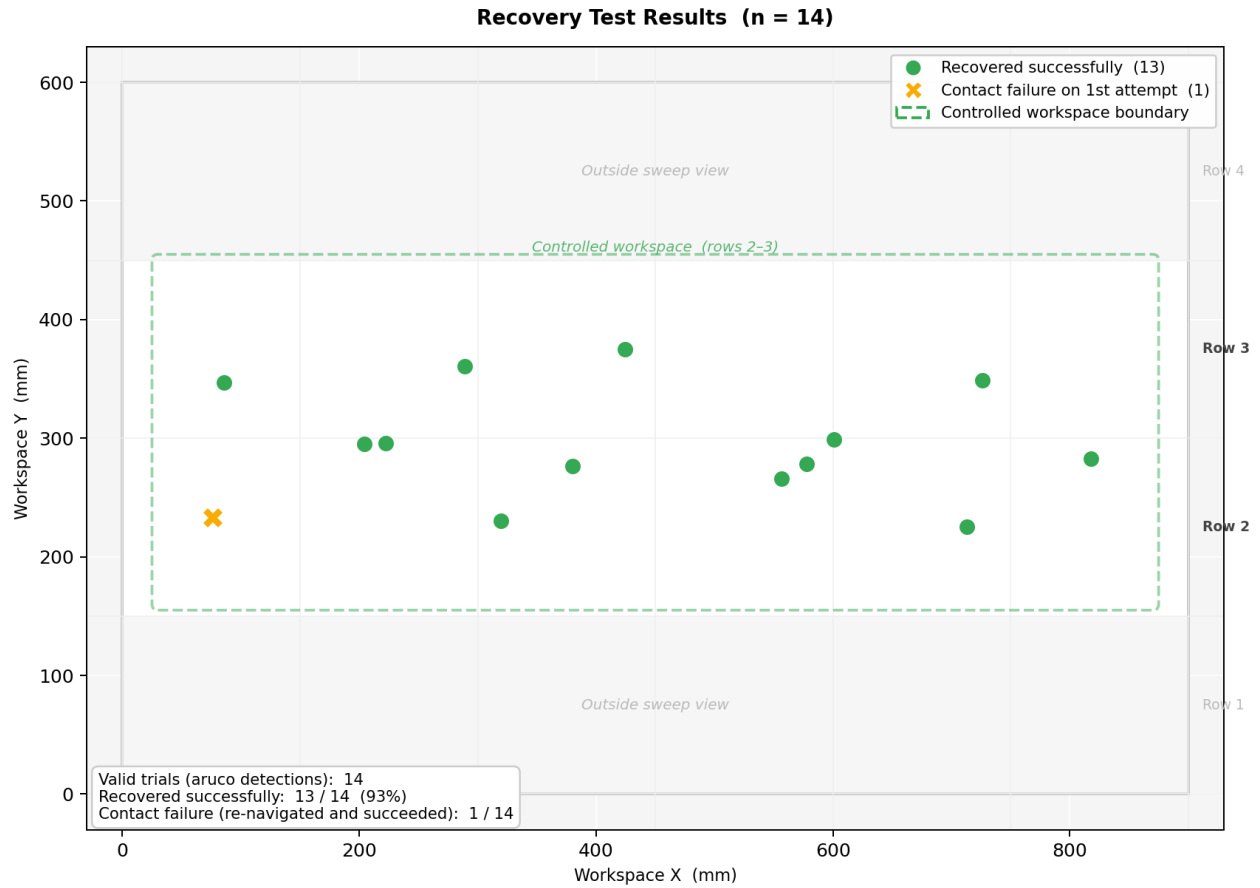


Figure 5.1d: Recovery test results showing hover positions across 14 trials

The single unsuccessful trial on the first attempt was a contact failure caused by a lateral hover offset, although the search sweep and re-navigation succeeded, the pose estimation error at that position was large enough to prevent a secure grasp on the first attempt. The system retried and succeeded on the second attempt. No trial within the tested workspace resulted in the object being undetectable during the search sweep.

5.2 Performance Analysis

Table 7 summarises the outcome of each objective against its defined target.

Objective	Metric	Target	Result	Status
Obj 1: Grasp Failure Detection	Accuracy / Recall / F1	$\geq 90\%$ accuracy	92% / 100% / 0.92	✓ Met
Obj 2: Autonomous Recovery	Recovery success rate	$\geq 80\%$	93% (13/14)	✓ Met
Obj 3: Navigation	Hover alignment rate	Consistent across workspace	92% (11/12)	✓ Met
Obj 4: Grasp Maximisation	Qualitative	Proof of concept	Demonstrated via ArUco	✓ Partial
Obj 5: Slip Detection (Extension)	Detection during transit	Object loss detected	Verified by pipeline design	✓ Met

Table 7: Summary of results against objectives and targets

All quantitative objectives were met or exceeded. The classifier accuracy of 92 percent exceeded the 90 percent target, and the 100 percent recall on the holding class confirms no genuine grasp was misclassified as empty across all 50 trials. A false negative in this context would cause the system to discard a successfully grasped object, making it the more operationally damaging error type. The recovery success rate of 93 percent exceeded the 80 percent target, and the navigation accuracy of 92 percent was consistent across the tested workspace.

The four false positives from the classifier are an expected artefact of the close-range viewpoint rather than a classifier failure. The pipeline design eliminates this condition by classifying only at a 400 mm clearance height, where no false positives were observed, directly validating that design decision.

The navigation failure at P1 is consistent with the known behaviour of ArUco PnP, lateral error in pose estimation increases with distance from the robot centre, and P1 sits at the far outer edge of the tested workspace. All 11 other positions produced successful grasps, confirming the system is reliable across the core workspace area.

Overall, all five objectives were either fully met or partially met. Objectives 1, 2, and 3 were fully met, with all three exceeding their quantitative targets. Objective 5 was met through pipeline design and verified through system behaviour during testing. Objective 4 was partially met; the gripper rotation mechanism and vertical descent approach were implemented and functional, but orientation estimation for markerless objects was not completed within the project timeline, leaving the full objective as a transferable proof of concept using ArUco-based alignment.

5.3 Discussion

The two-layer detection architecture performed as intended. Layer 1 provides an immediate hardware-level check that catches complete misses without performing an unnecessary lift. Layer 2 resolves the false positive ambiguity that Layer 1 alone cannot address. Together, they cover all three failure modes identified in Section 1.2.

The depth-based fallback navigation method was implemented but not independently validated, as an unresolved error in the transformation chain produced unreliable results within the project timeline. The ArUco method proved sufficient for the test scope, but addressing the depth fallback would be necessary for generalising to environments where the marker may not always be visible.

Grasp maximisation via OBB-based markerless orientation estimation was not completed within the project timeline. The ArUco axis alignment serves as a proof of concept demonstrating that the full gripper rotation mechanism is functional. Extending this to markerless objects would complete the objective in a future iteration.

Transit slip monitoring was verified through pipeline design rather than independently tested. The three-frame confirmation mechanism is in place, and the classifier runs continuously during transit. A dedicated slip test with known object drops at different carry positions would provide empirical validation of this component.

The test scope in this project is limited in scale and controlled in conditions. 14 recovery trials and 12 navigation positions provide a reasonable indication of system behaviour but are not sufficient for a rigorous statistical evaluation. A more comprehensive evaluation would include a larger number of trials across a wider range of positions, stress testing at the edges of the workspace

where pose estimation error is highest, and edge case testing with varied object orientations, different lighting conditions, and partial marker occlusion. Testing the depth-based fallback navigation independently once the transformation chain error is resolved would also allow a more complete evaluation of the full navigation pipeline. Additionally, end-to-end full pipeline trials, running all six stages in sequence without direct stage invocation, would better reflect real deployment conditions and reveal failure interactions between stages that isolated component testing does not capture.

5.4 Comparison with Related Work

The classifier accuracy of 92 percent compares against Nair et al. [20], who report 97 percent per-frame accuracy for image-based grasp verification from a fixed external camera. The 5 percent gap is consistent with the increased difficulty of operating from a wrist-mounted eye-in-hand camera at varying heights and positions rather than a controlled fixed viewpoint. Despite this, the 92 percent result still exceeds the conservatively set 90 percent target and confirms the approach is viable for this hardware configuration.

The recovery success rate of 93 percent exceeds the 83 to 88 percent closed-loop re-grasping success rates reported by Morrison et al. [21] for comparable parallel-jaw gripper hardware. The test set in this project is smaller and conducted in a controlled workspace, so a direct comparison should be interpreted cautiously. The result nonetheless demonstrates that the recovery routine is effective within the tested conditions and competitive with published benchmarks on similar hardware.

Zhu et al. [2] demonstrate a 40.7 percent improvement in overall task efficiency through closed-loop failure detection and recovery across a full pick-and-place cycle on a soft gripper platform. The approach taken in this project achieves comparable task completion capability on a rigid parallel-jaw gripper without requiring the pneumatic pressure sensing that Zhu et al. rely on, using only binary contact signals and visual classification to cover the same failure modes.

Chapter 6: Conclusion and Future Work

6.1 Summary of Contributions

This project designed, implemented, and evaluated a closed-loop grasp failure detection and recovery system on a UR10 robot arm with an OnRobot RG2 rigid parallel-jaw gripper. The core contribution is a six-stage autonomous pick-and-place pipeline that integrates failure detection at three points in the execution cycle and executes autonomous recovery without human intervention when failure is confirmed.

The project demonstrates that reliable closed-loop failure detection and recovery is achievable on a rigid parallel-jaw gripper using only binary contact signals and visual classification, without requiring dedicated force-torque sensing, tactile arrays, or external camera infrastructure. The two-layer detection architecture resolves the false positive ambiguity that hardware signals alone cannot address, and the unified recovery routine handles all three identified failure modes from a single implementation. All quantitative objectives were exceeded, with the classifier and recovery results both surpassing their respective targets. The eye-in-hand camera configuration at 21 degrees downward tilt proved effective as a single fixed viewpoint supporting detection, navigation, verification, and slip monitoring throughout the full cycle without any additional check pose.

6.2 Limitations

The system currently depends on ArUco marker visibility for reliable navigation. The depth-based fallback was implemented but not validated due to an unresolved error in the transformation chain, limiting the system to marker-visible pick orientations for dependable pose estimation.

Testing was conducted in a controlled single-object workspace under consistent lighting conditions. The evaluation set is limited in scale; 12 navigation positions and 14 recovery trials provide a reasonable indication of behaviour but are insufficient for a rigorous statistical analysis. Performance under varied lighting, partial occlusion, cluttered scenes, or with novel object classes has not been evaluated.

Layer 1 detection relies solely on binary contact and jaw-width signals, which cannot distinguish between failure subtypes beyond the 11 mm threshold boundary. This limits the granularity of hardware-level failure classification and means ambiguous cases must

always be forwarded to Layer 2 for visual confirmation, adding a lift cycle even when a simpler signal could resolve the outcome.

The current abort condition triggers after five total failures regardless of type, with no mechanism to distinguish between transient failures where retrying is likely to succeed and systematic failures where the environment has changed beyond what recovery can address.

6.3 Future Work

The most immediate priority is resolving the transformation chain error in the depth-based fallback navigation to remove the dependency on marker visibility. Once reliable, the fallback should be evaluated under the same navigation test protocol used in this project to establish a baseline for markerless navigation performance.

Completing Objective 4 in full requires implementing oriented bounding box estimation to extract object orientation from the RGB-D point cloud without a physical marker. The gripper rotation mechanism is already functional and would apply immediately once a reliable orientation source is available.

A more rigorous evaluation would include a larger number of trials, stress testing at workspace boundaries where pose estimation error is highest, varied lighting and occlusion conditions, and full end-to-end pipeline trials running all six stages in sequence. This would provide a more complete picture of system robustness and reveal any failure interactions between stages that isolated component testing does not capture.

The 70 mm descent value used in Stage 3 was determined empirically through testing rather than derived systematically from gripper geometry and object dimensions. A more principled approach would calculate the optimal descent depth from the known gripper cushion length, TCP offset, and object height, and validate this analytically before testing. This would generalise the grasp depth calculation to different object sizes and gripper configurations without requiring manual tuning.

Integrating a force-torque sensor or motor current sensing would provide richer Layer 1 contact feedback, enabling finer-grained failure classification and potentially reducing reliance on Layer 2 visual confirmation for straightforward miss cases. Generalising the detection and classification models to multiple object classes by retraining on a broader dataset would also extend the system beyond the current single-object scope.

Extending the pipeline to support continuous multi-cycle operation would be a natural next step for real deployment. Currently, the system completes a single pick-and-place cycle

and returns to the scan pose. A looped pipeline would allow the robot to autonomously repeat the full cycle, picking and placing multiple objects in sequence without any manual reset between cycles. This would require managing the drop zone state across cycles and handling scenarios where previously placed objects may re-enter the workspace.

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Appendix

Appendix A: Blog Post URL

Find here: <https://myuwaishin.github.io/final-year-blog/>

Find final demonstration video here: [Final Demo Video](#)

Appendix B: Risk Assessment

PDE3823 Risk Assessment Template 2025-26										Date:		17-09-25						
Project Title	Grasp Failure Detection and Recovery in Robotic Arm Pick-and-Place		Event Name:		Individual Project			AT RISK		RISK LIKELIHOOD		RISK SEVERITY						
Project Owner:	Myu Wai Shin		Site Supervisors:					Contractors	CON	Very Likely		Death						
Stand Number:	M00964135		Start Date:		18-Jan-26			Organisers, Partners & Stakeholders	OPS	Likely		Major Injury						
Stand Size (sqm):			End Date:		17-Apr-26					Possible		3 Day + Injury						
Stand Type:			Responsible Person:		Myu Wai Shin (student)			Visitors & Public	VP	Unlikely		Minor Injury						
Telephone:			E-mail:		ms3433@live.mdx.ac.uk			All persons onsite	APO	Very Unlikely		Near Miss						
TASK / ACTIVITY	HAZARD		WHO MIGHT BE HARMED		RISK LIKELIHOOD		RISK SEVERITY		RISK LEVEL		CONTROL MEASURES PUT IN PLACE TO REDUCE RISK		NEW RISK LIKELIHOOD		NEW RISK SEVERITY		CONTROLLED RISK LEVEL	
UR10 robot arm operation during pick-and-place	Collision between moving arm and operator		All persons onsite		Possible		3	Minor Injury	2	6	Maintain a safe distance during execution. E-Stop kept within immediate reach at all times. Operate at a low speed during testing. Safety induction completed.		Unlikely		2	Minor Injury	2	6
Gripper closing during manual object placement	Finger/hand caught between closing gripper jaws		All persons onsite		Possible		3	Minor Injury	2	6	Gripper only activated via command. RG2 max grip force is 40N - low risk of serious injury. Avoid placing your hand between the grippers.		Unlikely		2	Minor Injury	2	4
RG2 gripper operations	Gripper overheating or mechanical failure.				Possible		3	Minor Injury	2	10	Gripper monitored for unusual sounds or heat.		Possible		3	Minor Injury	2	6
Object pick and place	Object dropped or ejected from gripper at speed. Slip or trip from scattered objects.		All persons onsite		Likely		4	Minor Injury	2	6	Test objects are lightweight. Avoid staying within the pick-and-place workspace during execution.		Possible		3	Minor Injury	2	6
Robot moving to programmed pick positions	Arm sweeping through unexpected paths or colliding with obstacles.		All persons onsite		Possible		3	Minor Injury	2	6	Verify safe trajectory in each run. Home position verified to be a safe, retracted configuration.		Possible		3	Near Miss	1	3
Mounting the camera eye-in-hand	Camera drop or damage due to an insecure mount or a fast robotic arm motion.		All persons onsite		Possible		3	Minor Injury	2	10	Operate robotic arm within a low-speed range and ensure the camera is securely mounted before testing.		Unlikely		2	Minor Injury	2	4
Code and data management	Code or trained model data loss.				Possible		3	Near Miss	1	12	All code under Git version control. Datasets and models auto-synced to OneDrive. Blog serves as ongoing experiment log.		Very Unlikely		1	Near Miss	1	1
YOLO model inference / pose estimation pipeline	Model misclassification causing incorrect grasp target or incorrect approach pose sent to robot, causing collisions or unnecessary movements.		All persons onsite		Likely		4	Minor Injury	2	6	Only fire high-confidence detections. Poses sanity-checked against known workspace bounds before sending to robot. "No pick" safe skip if detection confidence is low.		Possible		3	Minor Injury	2	6
Project delivery	Recovery logic implementation too complex, insufficient time to complete before submission deadline				Possible		3	Near Miss	1	3	Defined three contingency plans (A/B/C). Plan B (failure detection + retry recovery) is the minimum viable deliverable.		Possible		3	Near Miss	1	3

Find here: [Project Risk Assessment Myu M00964135.xlsx](#)



Generative AI Use Disclaimer Form

This form is intended for PDE3823 students submitting their final-year project reports. Please complete the checklist below to provide transparency regarding the use of Generative AI tools during the project work. Submit this form alongside your final report.

Student Details:

- **Name:** _____ Myu Wai Shin _____
- **Student ID:** _____ M00964135 _____
- **Project Title:** _____ Grasp Failure Detection & Recovery in Robotic Arm Pick-and-Place System _____
- **Supervisor:** _____ Inas El-Aroussi _____

Generative AI Usage Declaration

Please indicate how generative AI tools were used during your project by ticking the relevant boxes below:

- ☐ No Generative AI tools were used during the completion of this project.
- ☒ AI tools were used for brainstorming and idea generation.
- ☒ AI tools were used for text drafting and report writing assistance.
- ☒ AI tools were used for code generation, debugging, or suggestions.
- ☒ AI tools were used for data analysis or visualisation support.
- ☐ AI tools were used for design simulation, modelling, or testing.
- ☐ Other (Please specify): _____

Acknowledgment

I hereby confirm that:

- All AI-generated content used in this report was critically evaluated, edited, and verified for accuracy and appropriateness.
- All intellectual contributions remain my own, and the final report reflects my independent work and understanding.
- Proper acknowledgment has been given where applicable.

Student Signature: _____  **Date:** _____ 15/04/2026 _____



UR10

Technical details

Specification

Payload	10 kg /22 lbs
Reach	1300 mm / 51.2 in
Degrees of freedom	6 rotating joints
Programming	Polyscope graphical user interface on 12 inch touchscreen with mounting

Performance

Repeatability	±0.1 mm / ±0.0039 in (4 mils)
Ambient temperature range	0-50°C
Power consumption	Min 90W, Typical 250W, Max 500W
Collaboration operation	15 advanced adjustable safety functions. TÜV NORD Approved Safety Function Tested in accordance with: EN ISO 13849:2008 PL d

Movement

Axis movement robot arm	Working range	Maximum speed
Base	± 360°	± 120°/Sec.
Shoulder	± 360°	± 120°/Sec.
Elbow	± 360°	± 180°/Sec.
Wrist 1	± 360°	± 180°/Sec.
Wrist 2	± 360°	± 180°/Sec.
Wrist 3	± 360°	± 180°/Sec.
Typical tool	1 m/Sec. / 39.4 in/Sec.	

Features

IP classification	IP54
ISO Class Cleanroom	5
Noise	72dB(A)
Robot mounting	Any
Estimated life at rated loads	35,000 hours
I/O ports	Digital in 2 Digital out 2 Analog in 2 Analog out 0
I/O power supply in tool	12 V/24 V 600 mA in tool

Physical

Footprint	Ø 190mm
Materials	Aluminium, PP plastics
Tool connector type	M8
Cable length robot arm	6 m / 236 in
Weight with cable	28,9 kg / 63.7 lbs

Universal Robots A/S
Energivej 25
DK-5260 Odense S
Denmark
+45 89 93 89 89

www.universal-robots.com
sales@universal-robots.com

Control Box

Features

IP classification	IP20
ISO Class Cleanroom	6
Ambient temperature range	0-50°C
I/O ports	Digital in 16 Digital out 16 Analog in 2 Analog out 2
I/O power supply	24V 2A
Communication	TCP/IP 100Mbit, Modbus TCP, Profinet, EthernetIP
Power source	100-240 VAC, 50-60 Hz

Physical

Control box size (WxHxD)	475 x 423 x 268 mm 18.7 x 16.7 x 10.6 in
Weight	15 kg / 33.1 lbs
Materials	Steel

Teach Pendant

Features

IP classification	IP20
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Physical

Materials	Aluminium, PP
Weight	1,5 kg / 3.3 lbs
Cable length	4,5 m / 177 in



Appendix E: Gripper Feedback: DI8 and AI2

I/O Feedback

The gripper uses DI8, DI9 and AI3 to give feedback on its status and finger position.

Use DI8 to detect if the gripper grabbed a workpiece or stopped at a given position.

DI9 will go LO (Busy) when the gripper is programmed, moving or changing force.

Digital status Feedback	Tool Inputs	
UR Version 3	0	1
UR Version 1 & 2	DI8	DI9
Position Reached	LO	-
Force Reached	HI	-
Gripper Busy	-	LO
Gripper Ready	-	HI

Analog feedback

AI2 outputs a voltage corresponding to the gripper width.

Analog Feedback			
	UR Input	Voltage	Width
Actual Width @ 0V:5V	AI2	0...3.7V*	0...110mm
Actual Width @ 0V:10V	AI2	0...3.0V*	0...110mm

* Due to the gripper analog output resistance (10kΩ), the analog feedback voltage will be affected by the robot input resistance. For the robots from Universal Robots, the input resistance is 29kΩ @ 0V:5V and 15kΩ @ 0V:10V, that result in a maximum input voltage of $5V \cdot \frac{29k\Omega}{10k\Omega + 29k\Omega} = 3.7V$ @ 0V: 5V and

$5V \cdot \frac{15k\Omega}{10k\Omega + 15k\Omega} = 3.0V$ @ 0V: 10V which correspond to the maximum width of 110mm.

The actual width can be calculated by $\frac{\text{voltage}}{\text{maximum input voltage}} \cdot 110\text{mm}$.

Appendix F: Camera Intrinsic Formula

$$X_{\text{cam}} = (u - cx) \cdot Z / fx$$

$$Y_{\text{cam}} = (v - cy) \cdot Z / fy$$

$$Z_{\text{cam}} = Z_{\text{smoothed}}$$